Like this course? Become an ex

Course Home (/learn/analytics-excel/home/welcome) > Week 6 (/learn/analytics-excel/home/week/6) > Case Study: Modeling Credit Card ...

Review Classmates: Part 5: Modeling Credit Card Default Risk and Customer Profitability

Review by December 16, 11:59 PM PT

Reviews 3 left to complete

Final Assignment

by Rajesh Radhakrishnan
December 15, 2015

like Flag this submission

What was the binary classification model you developed in Final Quiz Part 1, Question #1? Type your model in the box below.

Model = -1.12057478661328*(DEBT-INCOME-RATIO)-0.0343062*(AVERAGE of years at EMPL and Current ADDR) -0.0002324*age+0.18865038

AUC (on training set) = 0.81
AUC (on test set) = .79

Please refer to the Discussion for Peer Assessment document (https://d396qusza40orc.cloudfront.net/phoenixassets/Jackie%27s%20folder/Discussion%20for%20Peer%20Assessment.pdf) and the Peer Review Comments and Discussion (https://d396qusza40orc.cloudfront.net/phoenixassets/Jackie%27s%20folder/Final_Peer-Review-Discussion_10-30-2015.pdf) document during the review process.

Does the submitted model include at least 2 of the 6 pieces of data about the credit card applicants (age, years at current employer, years at current address, income over the past year, current credit card debt, and current automobile debt)?

- (selected) 2 pts
  Yes
- 1 pt
  No, the model only includes 1 of the pieces of data
- 0 pts
  No model was provided, or the model includes none of the pieces of data

Help Center

At the threshold score for the training set, what is the cost per event on the test set? Refer back to Final Project Quiz 1, question 6 for your answer.

Note: if you find that your costs per event on the test set are much higher than your costs per event on the training set, you should consider making your model simpler – probably using fewer input variables – as it is probably still over-fitting the training set data. Problems that are were not obvious at the ROC-curve stage may emerge when minimizing costs.

Cost per event = $938

@threshold =0.25

Is the minimum cost per event submitted by your peer an acceptable cost per event?

- 0 pts
  Cost per event > $1,250 (no incremental value)
- 1 pt
  Cost per event > $1,000 to <=$1,250
- (selected) 2 pts
  Cost per event >$800 to <= $1,000
- 3 pts
  Cost per event $600 to <=$800
- 0 pts
  Cost per event <$600 (unrealistic to expect that any model could produce a cost per event this low)

Your boss asks you to recommend a standard procedure for selecting future credit card applicants. Based on internal data alone (*not using the Eggertopia scores*) and the test set performance of your two models (the binary classification model from Final Project Quiz 1 and the linear regression model from Final Project Quiz 4), which method would you recommend - binary classification, linear regression or some combination or modification of the two? Explain why you recommended what you did, referencing at least two topics/points from the course.

This is a typical classification problem. Multivariate linear regression is not the best fit for classification problems. It is very appealing to run a multivariate regression by keeping the response variable as profitability score. However, this could potentially lead to several False Negatives as dependent variables (only internal data) used are not good enough to capture long-term behavioral aspect of customers. A logistic regression (again a classification approach) would be more appropriate than the binary classification.

Does the learner provide a clear recommendation for which model (or combinations of models) to use to select future credit card applicants?

- (•) 2 pts
  Yes, there is a clear recommendation
- ( ) 1 pt
  There seems to be a vague recommendation, but it is not clearly or definitively stated
- ( ) 0 pts
  No, there is no recommendation at all

No matter which model(s) is recommended, does the learner provide an explanation for their choice which references at least two topics/points from the course?

- ( ) 3 pts
  Yes, the learner mentions at least two topics/points from the course, and includes the idea that the best model is the one with the lowest cost per event
- ( ) 2 pts
  Yes, the learner explains and mentions at least two topics/points from the course, although does not explicitly mention that the best model is the one with the lowest cost per event
- (•) 1 pt
  No, the learner only references one topic/point from the course
- ( ) 0 pts
  No, the learner provides no explanation or her/his explanation has no connection to the course materials

Submit Review

Comments
Visible to classmates

share your thoughts...

Rajesh Radhakrishnan 2 minutes ago

correction: cost per event was $813 (AUC=.81) on test data and $688 (AUC=.79) on the training set.

(/learn/analytics-excel/supplement/XlBjU/exact-answers-for-final-project-quiz-4)
(/learn/analytics-excel/supplement/szSvv/feedback-survey)
(https://accounts.coursera.org/nerds/coursera/help?return_to=https://learner.coursera.help/hc)